

RoboTrack Platform

Example Technical Specification

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Eng. Divyansh Jha & AI Assistant

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Abstract

This is a working demo document compiled via the **Latex AI Workflow** engine. It demonstrates math equation rendering, code syntax highlighting, and responsive table formatting without float bottlenecks.

1 Core Differential Kinematics

The circular-arc state estimation model is formulated as:

$$\Delta s = \frac{\Delta s_L + \Delta s_R}{2}, \quad \Delta \theta = \frac{\Delta s_R - \Delta s_L}{W} \quad (1)$$

$$x_{k+1} = x_k + \frac{\Delta s}{\Delta \theta} (\sin(\theta_k + \Delta \theta) - \sin \theta_k) \quad (2)$$

2 Interface Control Specification

Topic	Type	Rate	Description
/odom	nav_msgs/Odometry	100 Hz	Filtered 6-DoF state estimate
/pose	geometry_msgs/PoseStamped	100 Hz	Front-face reference pose
/imu	sensor_msgs/Imu	100 Hz	Calibrated accelerometer and gyro telemetry

Table 1: Interface Definition Table